Oliver Perrin

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Experience

Machine Learning Engineer (Founding Team) May 2026 – Present
Stealth Startup *Remote*

- Sole ML engineer on the four-person founding team, responsible for ML research, model development, evaluation, inference, and application integration. Joined through a university capstone collaboration and continued into a full-time role.
- Build the infrastructure around model development: data preparation and validation, reproducible experiments, evaluation pipelines, and regression checks for release decisions.
- Trace failures across data, model outputs, APIs, and application behavior; develop targeted tests to distinguish modeling errors from integration defects.
- Own ML architecture and the supporting backend and data infrastructure, working directly with the founding team to turn research into reliable application behavior.

Founder Mar 2026 – Present
LiminalML *Remote*

- Founded LiminalML, a SaaS company for learning and explaining machine learning. Independently own product design, engineering, deployment, and subscription billing.
- Built Study for guided learning through intuition, mathematics, implementation, and recall; built Studio for creating, editing, sharing, and embedding interactive technical visuals.
- Develop the full stack with Next.js, TypeScript, Supabase, and Stripe. Write technical essays on transformer architecture, tokenization, and implementation tradeoffs.

Systems Engineering Intern June 2023 – August 2024
Triaxis Power Consulting LLC *Charlotte, NC*

- Built a Raspberry Pi monitoring system with thermocouples and potentiometers to measure pipe temperature and thermal expansion in power-plant equipment.

Research & Open Source

LexiMind | *Independent NLP research* Nov 2025 – Present

- Implemented a transformer in PyTorch, including attention, encoder/decoder layers, relative-position bias, cached generation, and task heads; transferred pretrained weights into the custom implementation.
- Developing a study of joint adaptation, task specialists, and model merging for summarization and classification under comparable training budgets, with book-aligned supervision and parameter-efficient adaptation.

Selected software | *Open source* Projects & source

- **PlayAxis:** Built a React/FastAPI platform combining sports feeds, mapped event discovery, private training logs, goals, and community features with persistent user data.
- **SartheAtlas:** Built a Le Mans car and circuit explorer with React, TypeScript, Three.js, and MapLibre, combining interactive engineering studies with aerial imagery, terrain, and elevation profiles.
- **Pulse:** Built a local-first Electron/Node.js app and menu-bar dashboard for coding-agent usage, combining local transcripts and CLI metadata with subscription allowances and reset times.

Education

Appalachian State University May 2026
B.S. Computer Science, Minor in Mathematics, Data Science Certificate *Boone, NC*

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, C/C++, Java
Machine Learning: PyTorch, Transformers, multi-task learning, fine-tuning, LoRA, model evaluation
Engineering: FastAPI, React, Next.js, Node.js, Docker, MongoDB, PostgreSQL/Supabase, CI/CD